
Is perceptual diminishing returns channel-organized, and specifically cone-aligned? A model-comparison follow-up to “The non-Riemannian nature of perceptual color space”

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Abstract

Bujack et al. (2022) showed that human color-difference perception exhibits *diminishing returns* (DR) — large differences are perceived as less than the sum of their parts — on the achromatic axis, and Stark et al. (2025) showed DR on two chromatic lines whose strengths are mutually inconsistent with any single global scaling. The same authors conjectured that DR may decompose into per-cone nonlinearities, $D = f_L(\Delta L) + f_M(\Delta M) + f_S(\Delta S)$, but reported that the hypothesis “cannot be tested on measurements from two geodesics only.” We test the strongest version of that hypothesis supported by existing public data. Placing the Helm (1964) and Ekman (1954) hue-circle dissimilarity sets into a stated, normalized CIE 2006 LMS cone space, we compare a nested family — additive (M0), isotropic-DR (M1), per-cone DR (M2) — against a matched panel of five fixed non-cone bases (opponent, sRGB, CIE RGB, CIELAB, CIE LCh; M3), using held-out likelihood with selection-aware bootstrap confidence intervals. We first replicate the 2022 gray-axis DR finding on two independent datasets (crowd-sourced 2AFC and Wiebel et al. 2017 MLDS; fitted compression ratios and exponents match the original report). On the chromatic data we find a powered, specification-robust *negative* result: per-cone DR never improves held-out prediction over the isotropic model, and is always beaten by at least one matched non-cone control (Helm: LCh; Ekman: the post-receptoral opponent basis). A channel-organized achromatic-vs- S split is likewise beaten by its matched opponent control. Synthetic recovery shows the comparison *is* capable of detecting cone alignment at these sample sizes were it present (correct cone-verdict rate 87–98%), so the null is earned, not merely underpowered. Diminishing returns is therefore not compactly cone-organized on the sampled hue circles; if it is channel-organized at all, the receiving channels are post-receptoral, not the cones.

1 Introduction

For over a century, perceptual color space has been modeled as a Riemannian manifold: the perceived difference between two colors is the length of the geodesic connecting them. Bujack, Teti, Miller, Caffrey and Turton Bujack et al. [2022a] showed that this picture fails on the neutral axis: for points A, B, C in that order on the axis, perceived distances obey the strict subadditivity $D(A, B) + D(B, C) > D(A, C)$ (*diminishing returns*, DR), which no Riemannian metric can produce along a geodesic. Stark, Turton and Bujack Stark et al. [2025] extended the triad methodology to two chromatic lines at $L^*=70$ (green–pink and blue–orange), confirmed DR on both, and — crucially — showed that the DR strength is *not constant across color space*: a scaling fit on one line degrades predictions on the other. This rules out the simplest non-Riemannian correction, a single

location-independent scaling of a Riemannian metric, and poses the structural question we address here: *how is the nonlinearity organized?*

The strongest organizational hypothesis available is the one the original authors themselves raised and declared unstable with their data Stark et al. [2025]: that DR decomposes into *per-cone* contributions,

$$D(i, j) = f\left(f_L(\Delta L_{ij}) + f_M(\Delta M_{ij}) + f_S(\Delta S_{ij})\right), \quad (1)$$

with $\Delta L, \Delta M, \Delta S$ the cone-excitation differences and f_L, f_M, f_S, f monotone scalings. Testing Eq. (1) requires many directions through cone space; the two chromatic geodesics of Stark et al. [2025] provide two. The public hue-circle dissimilarity sets of Helm [1964] and Ekman [1954] sample directions densely around two near-isoluminant hue circles and, placed in a normalized cone space, provide enough directions to run the test.

Why the answer matters. This is not a curve-fitting exercise; the answer changes what is predictable and what is mechanistically plausible, and we state this *before* the result. (1) *Compact prediction.* If Eq. (1) holds, DR everywhere in color space is generated by three one-dimensional functions plus an outer scaling — a handful of measurements would predict DR along any locus, instead of an atlas measured direction by direction. (2) *Mechanism.* If DR is aligned to the cones, the compressive nonlinearity acts on each photoreceptor channel *before* the channels are combined (pre-combination). If it instead aligns to derived post-receptoral quantities (the opponent channels $L-M$, $S-(L+M)$, $L+M$), the compression acts *after* aggregation — a different site in the pathway. The granularity question (a full $L/M/S$ split vs. a coarser achromatic-vs- S split) is the empirical face of this distinction. (3) *What a null means.* If the cone model is no better than standard non-cone bases, DR is neither cone-organized nor cleanly channel-localized on any of these axes, and the visual system’s compression must be organized altogether differently — an informative outcome, which is why the cone hypothesis must be tested against *several* popular non-cone bases, not one. We emphasize that these stakes are asserted *conditionally*: our result is negative, and the mechanistic reading is correspondingly limited to what a powered null supports (Section 5).

Positioning against prior work. Our contribution begins exactly where Stark et al. [2025] stops. (a) The 2022 result Bujack et al. [2022a] is DR on the gray axis; we *replicate* it as a pipeline anchor but claim no novelty there. (b) “Now in Color” Stark et al. [2025] established DR on two chromatic lines and its non-constancy; we take that as given. In particular, showing DR is merely *direction-dependent* (anisotropic) restates their non-constancy result and is *not* our claim. (c) The CGF geometry paper Bujack et al. [2025] formalizes the non-Riemannian metric; it contains no per-cone empirical test. Our novel object is *cone-alignment*: whether the per-axis nonlinearity is specifically organized on the cone axes, which requires beating matched non-cone anisotropic controls (M3), not merely the isotropic model (M1).

Contributions. (1) A documented, verified pipeline placing all provided stimuli into a normalized CIE 2006 LMS space, with a stated gauge fixing the outer/inner scaling ambiguity. (2) A replication of the 2022 gray-axis DR on the original 2AFC data and, independently, on the Wiebel et al. MLDS lightness data (suprathreshold scale). (3) A cross-validated comparison $M0 \subseteq M1 \subseteq M2$ against a five-member matched non-cone control panel M3, on two chromatic datasets and two LMS normalizations, with selection-aware bootstrap CIs. (4) An explicit identifiability analysis: collinearity diagnostics, synthetic parameter recovery, and a powered cone-vs-nearest-control (opponent) discrimination check. (5) A hue-locus curvature-confound control (sensitivity-ratio decomposition, an arc-length additive baseline, and an S -axis-only DR variant). (6) The headline answer: on these data DR is *not* cone-aligned, and not even channel-organized beyond isotropy; the claim is calibrated to the earned null.

2 Methodology

2.1 Cone space, normalization, and gauge

Stimuli to cone space. Each stimulus is specified in CIE 1931 chromaticity plus luminance factor (Helm: Table I of Helm [1964]; Ekman: spectral-locus chromaticities at equal luminance, see data SOURCE notes). We convert $xyY \rightarrow XYZ$, then $XYZ \rightarrow LMS$ by a single 3×3 matrix fit by least

squares between the two *provided* spectral tables (CIE 2006 10° cone fundamentals and CIE 1931 2° CMFs, both from cvrl.org Colour & Vision Research Laboratory [2024]), over their common wavelength range (390–830 nm). The fitted matrix reproduces the fundamentals of a monochromatic 555 nm light to 1.9% relative error (max over L, M) and has condition number 3.35; the reference white is CIE Illuminant C ($x=0.31006$, $y=0.31616$, $Y_n=100$), the white of the Helm colorimetry. Verification: the white maps to (0.5, 0.5, 0.5) under the primary normalization, and the CIE RGB control basis derived from the provided CMFs recovers the known luminance coefficients (1, 4.589, 0.060) vs. the reference (1, 4.5907, 0.0601). All converted stimulus coordinates are logged.

Normalization (gauge on cone scaling). Primary: per-cone Michaelis–Menten on white-referred responses, $c = r/(1 + r)$ with $r = R/R_w$ for $R \in \{L, M, S\}$, so the white maps to (0.5, 0.5, 0.5) and all coordinates lie in $[0, 1]$; this is the physiological analog of the CIELAB cube-root and regularizes the large raw S dynamic range. Sensitivity: pure von Kries contrast $r = R/R_w$. All headline models are run under *both* and reported in full.

Outer/inner gauge. Every model has the form $D = f(\sum_k f_k(|\Delta x_k|))$. Without a gauge convention, “is the nonlinearity per-cone?” is not well defined, because a global concave f with linear f_k is observationally close to a linear f with concave f_k . We fix: (i) the outer $f(u) = u^\rho$, a one-parameter family (concave iff $\rho < 1$), shared by all non-M0 models; (ii) each inner function lives in a fixed five-dimensional *concave power basis* on its axis’s training-fold range $x_{\text{ref},k}$: $B_0(x) = x/x_{\text{ref}}$ (linear) and $B_j(x) = (x/x_{\text{ref}})^{q_j}$, $q = (2/3, 1/2, 1/3, 1/5)$, with nonnegative coefficients $a_{kj} \geq 0$ (sums of concave functions are concave; the linear term keeps M0 exactly nested). The identity point is $a_k = (A_k, 0, 0, 0, 0)$. Under this gauge the only question is *where* nonlinearity lives: shared and outer (M1) or per-axis and inner (M2/M3).

2.2 The model family

Let $\Delta c = (\Delta L, \Delta M, \Delta S)$ be the normalized cone difference of a stimulus pair and $x_k = |\Delta c_k|$ (dissimilarity is symmetric).

- **M0 (additive null):** $D = \sum_c w_c |\Delta c|$, $w_c \geq 0$. City-block additive; no DR. It is an L_1 -additive metric, not itself a Riemannian (L_2) metric: the object of study is the outer/inner nonlinearity within the additive-aggregation family, and the L_1 vs. L_2 choice is a stated limitation.
- **M1 (isotropic DR):** $D = (\sum_c w_c |\Delta c|)^\rho$ — one global scaling; the correction Stark et al. [2025] showed insufficient across two lines.
- **M2 (per-cone DR):** $D = (f_L(\Delta L) + f_M(\Delta M) + f_S(\Delta S))^\rho$ with f_c in the concave power basis. $M0 \subseteq M1 \subseteq M2$ exactly (M1: $a_c = (w_c, 0, 0, 0, 0)$; M0: also $\rho=1$).
- **M3 (matched non-cone panel):** the identical functional form and parameter count as M2, on five fixed, disclosed bases of matched dimension: (i) **opponent** ($L-M$, $S-(L+M)$, $(L+M)/2$); (ii) **sRGB** (linear γ removed); (iii) **CIE RGB** (from the provided CMFs); (iv) **CIELAB** (per-axis inner functions on $(\Delta L^*, \Delta a^*, \Delta b^*)$, Illuminant C); (v) **CIE LCh** (per-axis inner functions on $(\Delta L^*, \Delta C^*, \Delta h)$ with the hue difference as arc length at the pair’s mean chroma — a fixed function of the pair, not a data-adaptive linearization). All bases are fixed transforms of the stimulus coordinates; no fitted rotations, which would nest M2 and invert the test. Every basis’s axes are scaled to unit RMS pair-wise difference over the dataset’s stimuli, so no control carries a conditioning advantage unrelated to alignment.
- **M2c / M3c (coarse channel split):** two-channel versions on $\{(L+M)/2, S\}$ (cone-derived) and its matched controls $\{(L+M)/2, S-(L+M)\}$ (opponent-derived) and $\{L^*, C^*\}$ (CIELAB), for the resolvable-granularity question.
- **M2-1ax (single-axis DR):** only one axis nonlinear (others linear), used for the curvature control (S -axis-only) and for attribution.

2.3 Observation models (per dataset; never pooled)

The datasets are on different response scales and are never pooled into one likelihood. *Gray 2AFC and Wiebel MLDS (triads)*: Thurstonian. For triad $(s; t_1, t_2)$ with perceptual map $\psi = g(\cdot)$, $P(t_1 \text{ more different}) = \Phi\left(\frac{f(|\psi_s - \psi_{t_1}|) - f(|\psi_s - \psi_{t_2}|)}{\sigma}\right)$. Two conventions are run and both reported: **(R)** the mirror of the reference pipeline of Bujack et al. [2022a] (bootstrap-conditional $g; f$ with free amplitude, $\sigma=1$), and **(P)** our canonical scale-matched convention (split-conditional $g; f$ shape-only with fixed unit secant, free σ), which makes M1 vs. M0 a matched-scale comparison. **(P)** is primary. *Helm / Ekman (pairwise dissimilarities)*: Gaussian observation model $\delta_{ij} \sim \mathcal{N}(D_{\text{model}}(i, j), \sigma^2)$ with σ profiled on the training fold; held-out metric is the Gaussian negative log-likelihood (NLL) per observation. Helm is fit on per-subject vectors normalized to unit RMS (fixed preprocessing), with subject-disjoint folds.

2.4 Fitting, cross-validation, and comparison protocol

All models are fit by Nelder–Mead maximum likelihood with multiple restarts (seeds logged); convergence is 100% across all reported fits. Gray data: 10 train/test splits (50 held-out responses per triad, response-disjoint), 50 bootstrap resamples per split. Helm: leave-one-subject-out over the 11 color-normal subjects/sessions (N6a/N6b treated as separate sessions). Ekman: 10 pair-disjoint folds over the 91 pairs. Chromatic bootstrap: 300 resamples of the training side (subjects for Helm, pairs for Ekman), 5-fold CV inside each resample (levels therefore differ slightly from the main CV; all quoted *deltas* are paired within protocol). The headline M2-vs-best-M3 delta is *selection-aware*: the best control (lowest held-out NLL) is re-picked inside every bootstrap resample; we also report the paired $\min_k (M2 - M3_k)$.

2.5 Identifiability and confound control

Collinearity. We report the correlation of $|\Delta L|, |\Delta M|, |\Delta S|$ across pairs, variance inflation factors, the design condition number, and the principal angles between the cone and opponent design spaces on the sampled directions. *Synthetic recovery (labeled synthetic only)*. Data are generated at the real stimulus coordinates under a known per-cone M2 (distinct concave f_L, f_M, f_S) and under the nearest control (opponent-basis M3), with noise matched to the real residual scale, and the full CV comparison is re-run (100 simulations per condition). We report how often the correct model wins — including the binding case, M2-vs-opponent discrimination. *Curvature confound*. On a hue circle, locus curvature makes chord distances underestimate arc lengths, mimicking DR. We control this in three ways (details in Section 3): (a) a sensitivity-ratio decomposition $r = \text{arc}_u / \text{chord}_u$ per pair in the within-axis metric space (within-axis fits are curvature-free by construction, since each axis’s pairs are collinear in their own coordinate); (b) a two-parameter arc-length additive baseline compared on held-out NLL; (c) the verdict recomputed with only the S -axis (radial/chromatic) inner function free — the only component curvature can mimic.

3 Experimental Setup

Datasets and licenses. (i) Gray-axis 2AFC: 91,896 triad judgments on 320 triads spanning the neutral L^* axis Bujack et al. [2022a], public release from the authors’ repository (BSD-3) Bujack et al. [2022b]. (ii) Wiebel, Aguilar & Maertens MLDS lightness data Wiebel et al. [2017a] (OSF, CC BY 4.0 Wiebel et al. [2017b]): 10 observers $\times$ 5 viewing contexts $\times$ 120 triads $\times$ 10 repeats over 10 reflectances. (iii) Helm (1964) Helm [1964]: 10 Munsell tiles (Value 6.0, Illuminant C colorimetry transcribed from Table I), 45 dissimilarity pairs $\times$ 16 subjects (11 color-normal sessions used; 4 color-deficient subjects excluded from the primary analysis), matrices redistributed via smacof (GPL-3) de Leeuw et al. [2024]. (iv) Ekman (1954) Ekman [1954]: 14 spectral stimuli, 91 pairwise dissimilarities ($1 - s$ from the similarity matrix, via smacof, GPL-3). Cone fundamentals and CMFs are from cvrl.org Colour & Vision Research Laboratory [2024] and are cited per cvrl’s terms. No other human data are used; synthetic data appear only in the identifiability analysis and are labeled.

Preprocessing and folds. Helm subject vectors are normalized to unit RMS (ratio scales are subject-specific units). Folds: Helm subject-disjoint; Ekman pair-disjoint; gray response-disjoint. The Wiebel response coding (which integer encodes “pair (s_2, s_3) more different”) was determined

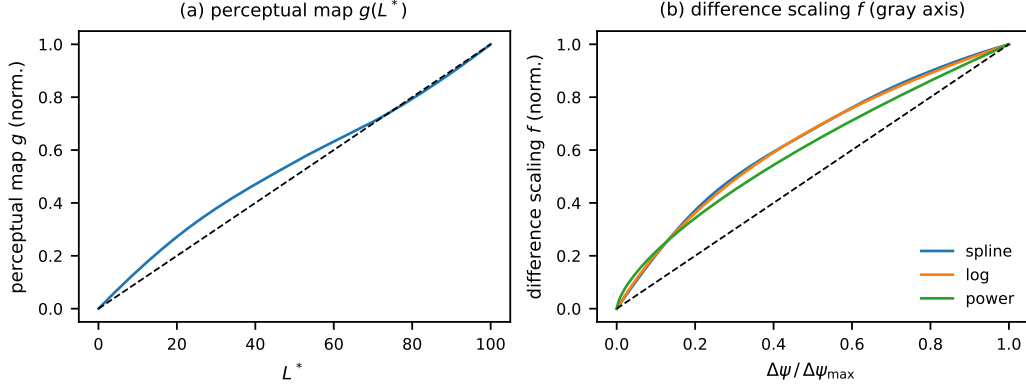


Figure 1: Gray-axis replication. (a) Perceptual map $g(L^*)$ (normalized to unit max), mean and 95% band over splits. (b) Difference-scaling $f(\Delta\psi)$ (normalized to unit secant), spline/log/power forms with 95% bands; curves above the diagonal are concave, i.e. diminishing returns.

from the data by an order-consistency check on positional extreme triads (fraction choosing the wide pair = 0.98, $n = 499$; reflectance-space correlation 0.58) and logged. The gray response coding was checked against the reference implementation’s convention ($R=1$ = test 2 chosen).

Held-out likelihood. For triad data, held-out Bernoulli NLL per response; for dissimilarity data, held-out Gaussian NLL per observation at the training-fold $\hat{\sigma}$. Identical folds for all models within a dataset.

4 Experimental Results

4.1 Gray-axis replication (anchor)

The 2022 finding replicates exactly in kind and closely in magnitude. The consensus regression (their Eq. 10) gives $\beta_{\bar{a}} = -0.0061$ ($t(28) = -5.73$, $p < 10^{-5}$, $R^2 = 0.73$; their Table 1: -0.005 , $t = -5.57$). Under the scale-matched convention (P), every concave- f model beats the identity- f M0 on held-out NLL (per-response reduction: power 0.0044 [0.0039, 0.0049], log 0.0056 [0.0050, 0.0061], sin 0.0042 [0.0037, 0.0047], spline 0.0058 [0.0053, 0.0063]; 95% CIs across folds), and under the reference convention (R) the ordering baseline $> g$ -only $> g+f$ reproduces with bootstrap CIs excluding zero (e.g. spline $g+f$ improvement 0.0058 [0.0044, 0.0076] nats/response). The fitted scaling is concave with compression ratio $f(0.5\Delta\psi_{\max})/f(\Delta\psi_{\max}) = 0.68$ (spline; 0.63 power; 0.68 log), against the paper’s $1.36/2.06 = 0.66$; the fitted log shape parameter $\hat{a} = 3.87$ (their highest-likelihood $a = 5.34$) and power exponent $\hat{a} = 0.67$. Figure 1 shows the perceptual map g and the scaling f with bands.

Independently, on the Wiebel MLDS data (suprathreshold reflectance axis), the concave-power model beats the identity model on held-out NLL in *all five* viewing contexts, with fitted exponents $a \in [0.32, 0.41]$ (all 95% CIs excluding 1; e.g. plain context $a = 0.341$ [0.330, 0.352], NLL improvement 0.0070 [0.0048, 0.0093] nats/judgment). DR on the achromatic axis is thus corroborated on an independent laboratory dataset.

4.2 Chromatic model comparison

Table 1 collects the headline numbers. The full held-out comparison (both datasets, primary normalization) is in Figure 2.

(a) Is DR direction-dependent at all (M2 vs M1)? No. On Helm, M2 is *worse* than M1 on held-out data: paired bootstrap delta $\Delta_{M2-M1} = -0.0638$ nats/obs (95% CI $[-0.0674, -0.0592]$; M2 better in 0/300 resamples). On Ekman the point estimate also favors M1, -0.0291 $[-0.0670, +0.0143]$, with M2 better in only 8.3% of resamples. Both normalizations agree (von

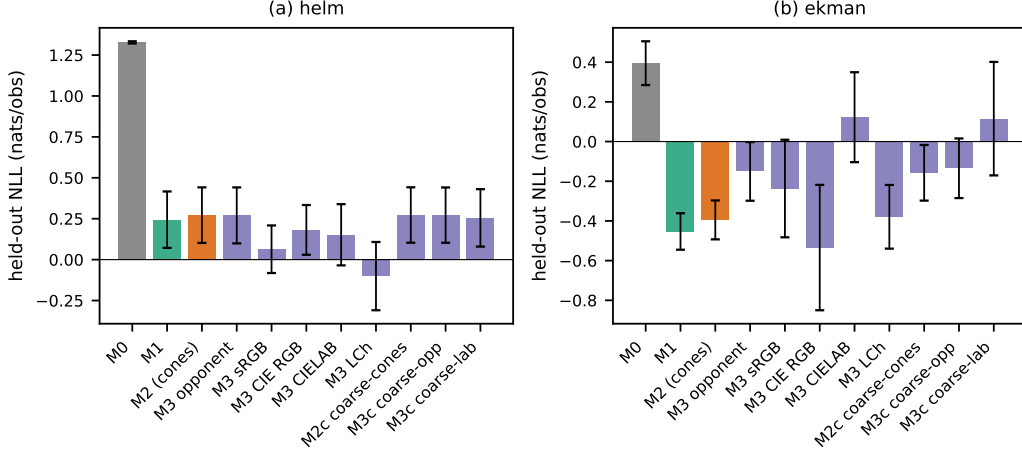


Figure 2: Held-out Gaussian NLL per observation (lower is better) for M0, M1, M2 and the full M3 non-cone panel, plus the coarse two-channel models; error bars are SD across folds. (a) Helm; (b) Ekman. Primary (Michaelis–Menten) normalization.

Kries: Helm -0.0797 $[-0.0831, -0.0753]$; Ekman -0.0633 $[-0.0927, -0.0361]$. M2’s extra parameters do not buy held-out improvement, and M2 does *not* collapse trivially to M1 (distinctness check below) — the flexibility is simply not supported. Meanwhile M1 beats M0 decisively on both datasets (Helm: $M0-M1 = -0.1732$ i.e. M1 better by 0.17 nats/obs, CI $[-0.1987, -0.1384]$; Ekman: -0.7834 $[-0.8866, -0.6498]$), so DR itself is present and strong.

(b) Is the structure cone-aligned (M2 vs best M3)? No. The selection-aware headline delta is negative under every specification: Helm (MM) best control is LCh in 300/300 resamples, $\Delta_{\text{bestM3-M2}} = -0.155$ $[-0.162, -0.147]$; Ekman (MM) best control is opponent in 297/300 resamples, -0.131 $[-0.233, -0.033]$. Under von Kries: Helm -0.180 (LCh, 300/300), Ekman -0.311 (opponent, 296/300). The cone basis beats only the weakest controls (sRGB, CIE RGB on Helm), never the best.

(c) Resolvable granularity. The coarse achromatic-vs- S model (M2c) does not beat its matched opponent control either: Helm $\Delta_{\text{coarse opp-coarse cones}} = -0.0199$ $[-0.0236, -0.0159]$; Ekman -0.0020 $[-0.0596, +0.0598]$ (indistinguishable). So even at the coarsest channel granularity, the data do not support cone-organized over opponent-organized structure.

Per-cone distinctness. The fitted M2 inner functions (Figure 3) are concave and non-degenerate: half-difference compression ratios $f_c(0.5 x_{\text{ref}})/f_c(x_{\text{ref}})$ on Helm: $f_L = 0.591$ $[0.516, 0.722]$, $f_M = 0.580$ $[0.523, 0.732]$, $f_S = 0.515$ $[0.501, 0.549]$; on Ekman: $f_L = 0.676$ $[0.512, 0.814]$, $f_M = 0.526$ $[0.505, 0.585]$, $f_S = 0.623$ $[0.523, 0.791]$ (bootstrap CIs). They are individually concave but the held-out data do not reward their per-axis freedom.

4.3 Identifiability: is the null earned?

Collinearity on the sampled directions is moderate, not extreme: $\text{corr}(|\Delta L|, |\Delta M|) = 0.41$ (Helm), 0.52 (Ekman); VIFs ≤ 2.5 ; design condition numbers 3.8 (Helm), 5.5 (Ekman). The cone and opponent design spaces differ substantially on these directions (largest principal angle 44° Helm, 51° Ekman) — the bases are distinguishable in principle. *Synthetic recovery* confirms the comparison is powered (Figure 4): with M2-cones as ground truth at the real coordinates and noise, M2 beats M1 in 81% (Helm) / 62% (Ekman) of simulations and beats the best non-cone control in 98% / 87% ; with opponent-M3 as ground truth, the opponent basis is correctly selected in $100/100$ simulations on both datasets and the cone model almost never wins spuriously (cone-alignment false-positive rate 0% Helm, 13% Ekman). The pipeline can therefore detect cone alignment at these sample sizes were it present — and on the real data it does not.

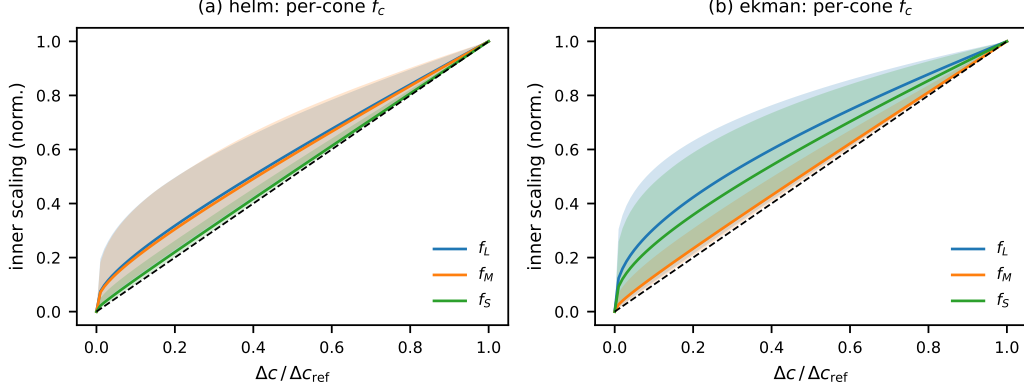


Figure 3: Fitted per-cone inner functions f_L, f_M, f_S (normalized to unit value at the axis reference difference) with 95% bootstrap bands. (a) Helm; (b) Ekman.

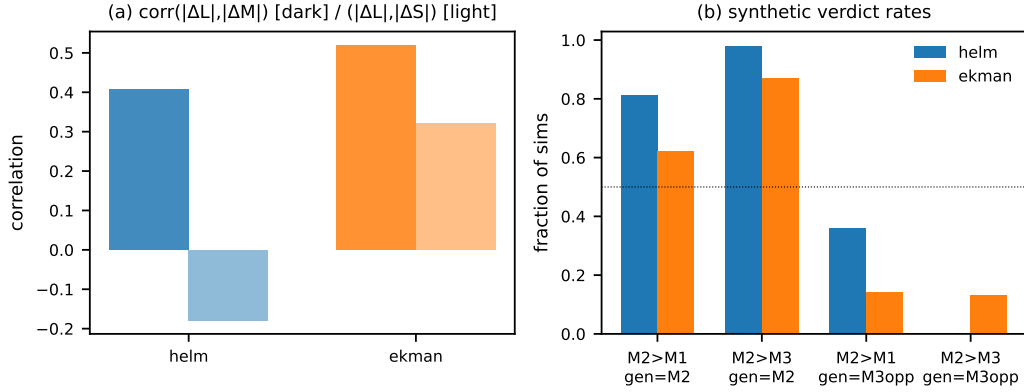


Figure 4: (a) Collinearity of cone differences on the sampled pairs. (b) Synthetic verdict rates: the CV pipeline recovers the true generating basis (M2-cones or M3-opponent) at the real sample sizes.

4.4 Curvature confound control

The sensitivity-ratio decomposition confirms the alias is real and must be controlled: within-axis-metric arc-to-chord ratios r exceed 1.1 for 78% (Helm) / 85% (Ekman) of pairs (median $r = 2.37 / 2.97$). However: (i) the two-parameter arc-length baseline is far worse than M1 on held-out NLL (Helm 0.0062 vs. M1 0.2441 — note lower NLL is better, and the arc baseline is beaten by M1 on Ekman too, -0.1389 vs. -0.4531), so curvature alone cannot explain the DR signal; (ii) the S -axis-only DR model (the only component curvature can mimic) does not beat M1 either (Helm $\Delta = +0.0005$; Ekman $\Delta = -0.0618$, i.e. worse); and (iii) the headline M2/M3 verdict is *unchanged* before and after restricting to the component free of the confound. The null is not a curvature artifact.

5 Analysis

Verdicts. (a) *Anisotropy*: not supported on these hue circles. M2 loses to M1 on held-out data in all four dataset $\times$ normalization cells (significantly in three; Ekman-MM inconclusive but negative in sign). (b) *Cone alignment*: rejected with power. The cone model is beaten by the best matched non-cone control in every cell (selection-aware CIs exclude zero in all four), and the synthetic discrimination check shows the test would have detected cone alignment 87–98% of the time were it present at the modeled effect size. (c) *Granularity*: the finest split the data can support is *no channel*

Table 1: Main results. Held-out NLL (nats/obs; lower is better) from the full cross-validation; deltas are paired, with 95% bootstrap CIs (selection-aware for the headline: the best M3 control is re-picked inside every resample). Primary (Michaelis–Menten) normalization; von Kries sensitivity agrees on every verdict. Negative delta values favor the second-named (comparison) model.

Quantity	Estimate	95% CI	Comparison
<i>Gray-axis DR replication (2AFC, 91,896 judgments; 320 triads)</i>			
M1–M0 held-out NLL (spline f)	−0.0058	[−0.0063, −0.0053]	2022: concave f
$f(0.5\Delta\psi_{\max})/f(\Delta\psi_{\max})$ (spline)	0.678	[0.673, 0.683]	2022: 1.36/2.06 = 0.66
Wiebel MLDS: M1–M0 (plain context)	−0.0070	[−0.0093, −0.0048]	suprathreshold
<i>Held-out fit, Helm (NLL/obs; 11-fold subject-disjoint CV, SD across folds)</i>			
M0 (additive)	1.3274	± 0.0067	
M1 (isotropic DR)	0.2441	± 0.1724	
M2 (per-cone DR)	0.2717	± 0.1697	
<i>Held-out fit, Helm, M3 panel (matched complexity)</i>			
M3 opponent	0.2703	± 0.1710	
M3 sRGB	0.0635	± 0.1454	
M3 CIE RGB	0.1816	± 0.1520	
M3 CIELAB	0.1518	± 0.1870	
M3 CIE LCh	−0.1008	± 0.2086	
<i>Held-out fit, Ekman (NLL/obs; 10-fold pair-disjoint CV, SD across folds)</i>			
M0 (additive)	0.3949	± 0.1102	
M1 (isotropic DR)	−0.4531	± 0.0920	
M2 (per-cone DR)	−0.3948	± 0.0981	
<i>Held-out fit, Ekman, M3 panel (matched complexity)</i>			
M3 opponent	−0.1505	± 0.1478	
M3 sRGB	−0.2367	± 0.2458	
M3 CIE RGB	−0.5343	± 0.3158	
M3 CIELAB	0.1227	± 0.2266	
M3 CIE LCh	−0.3792	± 0.1603	
<i>Headline deltas (paired bootstrap, 300 resamples; positive favors M2)</i>			
M2 – M1 (anisotropy), Helm	−0.0638	[−0.0674, −0.0592]	M1 wins
M2 – M1 (anisotropy), Ekman	−0.0291	[−0.0670, +0.0143]	n.s., favors M1
M2 – best M3, Helm (headline)	−0.1550	[−0.1621, −0.1468]	LCh wins, 300/300
M2 – best M3, Ekman (headline)	−0.1308	[−0.2333, −0.0325]	opponent wins, 297/300
<i>Identifiability</i>			
corr(ΔL , ΔM), Helm / Ekman	0.41 / 0.52	—	moderate, not extreme
Synthetic: M2 beats best M3 truth=M2	98% / 87%		Helm / Ekman
Synthetic: cone wins truth=opponent	0% / 13%		false-positive rate
Coarse M2c – M3c-opp, Helm / Ekman	−0.0199 / −0.0020	[−0.0236, −0.0159] / [−0.0596, +0.0598]	opponent control not better
<i>Hue-curvature confound control</i>			
Sensitivity ratio $r > 1.1$, Helm / Ekman	78% / 85%		median $r = 2.37 / 2.97$
Arc-length baseline NLL, Helm / Ekman	0.0062 / −0.1389		vs. M1: 0.2441 / −0.45
S -axis-only DR – M1, Helm / Ekman	+0.0005 / −0.0618		verdict unchanged after

split at all — isotropic DR (M1) is the best-supported DR structure on both hue circles; the coarse achromatic-vs- S split is not favored over its matched opponent control.

What the null does and does not say. The null is specific, not sweeping. It says: on two near-isoluminant hue circles, in a normalized CIE 2006 cone space, DR is not compactly generated by per-cone nonlinearities of the form Eq. (1), and no axis-aligned channel structure beats a single global scaling. It does *not* say DR is isotropic everywhere — Stark et al. [2025] already showed a single global scaling fails *across regions* (two lines at different chromatic locations). Reconciling the two: DR is *location*-dependent across color space Stark et al. [2025] but, within the sampled hue-circle neighborhoods, not *channel*-decomposable. The non-constancy they found lives in the spatial variation of the outer scaling, not in per-channel inner functions. The mechanistic reading is correspondingly scaled back: our data are consistent with the compression sitting *after* cone combination (or being spatially modulated), and inconsistent with a strong pre-combination per-

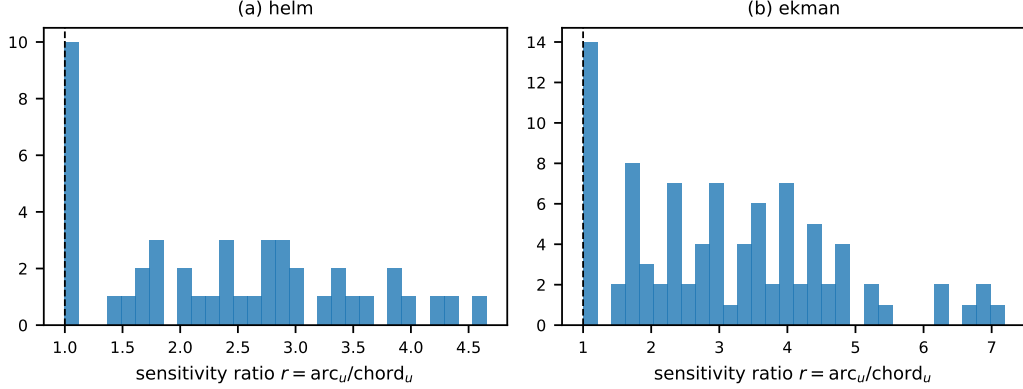


Figure 5: Distribution of the sensitivity ratio $r = \text{arc}_u / \text{chord}_u$ over pairs; $r=1$ (dashed) is the curvature-free case.

cone site; but because hue circles sample a limited subspace, this is evidence about the sampled region, not the whole space.

Failure analysis. Three regimes resist clean conclusions. (i) Ekman under MM normalization has the widest headline CI ($[-0.233, -0.033]$); the verdict is still negative but the margin is less stable, and the Ekman matrix is an aggregate over subjects with no per-subject uncertainty. (ii) The nearest-control discrimination, though powered at the modeled effect sizes, has finite sensitivity: a per-cone effect much weaker than the synthetic ground truth could hide (the synthetic margin, 0.016–0.05 nats/obs, is the scale the design resolves). (iii) On Helm, the achromatic within-axis fit for the curvature control rests on few achromatic-dominated pairs (1 of 45 at the 3:1 cutoff) and falls back to an all-pairs fit (logged); the Ekman control (19 A-dominated, 50 S-dominated pairs) and the arc-baseline comparison carry the weight.

6 Related Work

The Riemannian paradigm runs from Riemann through Helmholtz and Schrödinger (reviewed in Bujack et al. [2022a], Wyszecski and Stiles [1982]). Early evidence of non-additivity is due to MacAdam [1963] (polynomial exponent < 1), Helm [1964] (logarithmic transform), and Izmailov & Sokolov [1991] (spherical embedding); Bujack et al. [2022a] made the case decisive on the neutral axis with a Thurstonian difference-scaling MLE Maloney and Yang [2003], Knoblauch and Maloney [2008], Thurstone [1927]. Our replication uses their released data Bujack et al. [2022b] and an independent MLDS set Wiebel et al. [2017a,b]. The 2025 follow-ups are the direct antecedents: Stark et al. [2025] demonstrated chromatic DR and its non-constancy and posed the per-cone hypothesis as untestable on two geodesics; Bujack et al. [2025] formalized the non-Riemannian geometry (shortest paths in the perceptual metric vs. the induced Riemannian tensor) with no per-cone empirics. We add what both lack: a many-direction, cross-validated test of the per-cone decomposition against matched non-cone controls, with the identifiability analysis that the cone-vs- opponent discrimination requires. Our negative result is complementary to Stark et al. [2025]: their non-constancy says one global scaling cannot cover *different regions*; ours says the residual structure is not *cone-aligned channels* either. The hue-circle data themselves Helm [1964], Ekman [1954], de Leeuw et al. [2024], Borg and Groenen [2005] are classics of the MDS literature; we use them with explicit colorimetry rather than as bare similarity matrices.

7 Conclusion

On the public hue-circle data, perceptual diminishing returns is *not* channel-organized into per-cone components: the per-cone model never beats the isotropic model on held-out judgments, and always loses to at least one matched non-cone control (LCh on Helm, the post-receptoral opponent basis on Ekman). The null is earned: the gray-axis anchor replicates (twice, on independent datasets), the

identifiability analysis shows the test is powered to find cone alignment at these sample sizes, and the curvature confound is controlled and does not change the verdict. If DR is generated by channel-wise compression, the receiving channels are post-receptoral, not the cones; but the strongest defensible statement is weaker — within the sampled hue-circle regions, a single isotropic scaling is the best supported DR structure, and the cross-region non-constancy of Stark et al. [2025] must be carried by spatial variation of that scaling rather than by per-cone inner functions. *Limitations*: two hue circles at fixed lightness are a limited subspace; the Ekman data are subject-aggregated; the additive aggregation family is L_1 , not Riemannian L_2 ; and very weak per-cone effects below the demonstrated resolution cannot be excluded.

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